

ARMY INSTITUTE OF TECHNOLOGY

LAB MANUAL



Name of Subject	Lab Practices- III
Department	Information Technology
Subject Code	414446
TW/OR/PR	TW+OR
Marks	25+25 (Credits:02)
Course (Pattern)	2019
Year	BE
Semester	1

List of Experiments

(As given by SPPU)

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Pre-requisite for Lab

Companion Courses	
Code	Subject
414441	Information Storage and Retrieval
Prerequisite Courses/Lab	
Code	Subject
214443	Data Structures and Algorithms
214452	Database management systems

Learning Objectives

Sr. No	Objective
1	To understand the concepts of information retrieval.
2	To understand the role of clustering in information retrieval.
3	To study indexing structures for information retrieval.
4	To evaluate the performance of the IR system and understand user interfaces for searching.
5	To understand information sharing on the web.
6	To understand the various applications of information retrieval giving emphasis to multimedia and distributed IR, web Search.

Course Outcomes

By the end of the course, students should be able to,-

Sr. No	Course Outcomes
1	Understand the concept of Information retrieval with text analysis.
2	apply clustering in information retrieval
3	Evaluate performance of information retrieval systems.
4	Understand the concept of multimedia information retrieval.
5	Apply appropriate tools in analyzing the web information
6	Understand importance of recommender system with mapping of the concepts of the subject on recent developments in the Information retrieval field.

Precautions / Lab Safety Do's and Don'ts/ Lab Etiquettes

Sr. No	Details
1	The use of personal audio or video equipment is prohibited in the laboratory
2	Keep work area neat and free of any unnecessary objects.
3	Always perform the experiments as directed by your instructor.
4	Never work in the laboratory without the supervision of an instructor.
5	Mobile Phones are strictly prohibited in Laboratory
6	Keep your bags outside the laboratory.
7	Make entry in the Laboratory Logbook

Evaluation Guidelines (Rubrics)

(Similar guidelines pertaining to each subject (lab) shall be used to evaluate **each experiment** performed in the lab)

Grade	Poor	Average	Good	Outstanding
Marks	0-3	4-5	6-8	9-10

Criteria	Grade		Marks
Set-up and Equipment Care	Poor (0-3)	Set-up of equipment is not accurate, help is required with several major details	
	Average (4-5)	Set-up of equipment is generally workable with several details that need refinement	
	Good (6-8)	Set-up of equipment is generally accurate with 1 or 2 small details that need refinement	
	Outstanding (9-10)	All equipments accurately placed	
Following Procedure	Poor (0-3)	Lacks appropriate knowledge of the lab procedures. Often requires help from the teacher to even complete basic procedures	
	Average (4-5)	Demonstrates general knowledge of lab procedures. Requires help from the teacher with some steps in procedures.	
	Good (6-8)	Demonstrates good knowledge of the lab procedures. Will discuss with peers to solve problems in procedures.	
	Outstanding (9-10)	Demonstrates very good knowledge of the lab procedures. Gladly helps other students to follow procedures.	
Data Collection	Poor (0-3)	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	
	Average (4-5)	Measurements are somewhat inaccurate and imprecise. Observations are incomplete. There are 3 or more minor errors using symbols, units and significant digits or 2 major errors	
	Good (6-8)	Measurements are mostly accurate. Observations are generally complete. Work is organized. Only 2 or 3 minor errors using symbols, units and significant digits.	
	Outstanding	Measurements are both accurate and	

	(9-10)	precise. Observations are very thorough and may recognize possible errors in data collection. Work is neat and organized. Includes appropriate symbols, units and significant digits.	
Analysing and Concluding	Poor (0-3)	Provides limited analysis of the data. Demonstrates limited ability to draw conclusions based on the data.	
	Average (4-5)	Provides some analysis of the data. Demonstrates some ability to draw conclusions based on the data.	
	Good (6-8)	Provides sufficient analysis of the data. Draws valid conclusions based on the data.	
	Outstanding (9-10)	Provides rich analysis of the data. Draws insightful conclusions based on the data.	
Safety	Poor (0-3)	Proper safety precautions are consistently missed. Needs to be reminded often during the lab.	
	Average (4-5)	Proper safety precautions are often missed. Needs to be reminded more than once during the lab.	
	Good (6-8)	Proper safety precautions are generally used. Uses general reminders of safe practices independently.	
	Outstanding (9-10)	Proper safety precautions are consistently used. Consistently thinks ahead to ensure safety. Will often help other students to conduct labs safely.	
Specifications (Computer Program)	Poor (0-3)	The program is producing incorrect results.	
	Average (4-5)	The program produces correct results but does not display them correctly.	
	Good (6-8)	The program works and produces the correct results and displays them correctly. It also meets most of the other specifications.	
	Outstanding (9-10)	The program works and meets all of the specifications.	
Readability (Computer Program)	Poor (0-3)	The code is poorly organized and very difficult to understand.	
	Average (4-5)	The code is readable only by someone who knows what it is supposed to be doing	
	Good (6-8)	The code is fairly easy to understand.	
	Outstanding	The code is exceptionally well organized	

	(9-10)	and very easy to follow.	
Reusability (Computer Program)	Poor (0-3)	The code is not organized for reusability.	
	Average (4-5)	Some parts of the code could be reused in other programs.	
	Good (6-8)	Most of the code could be reused in other programs.	
	Outstanding (9-10)	The code could be reused as a whole, or each routine could be reused.	
Documentation (Computer Program)	Poor (0-3)	The documentation is simply comments embedded in the code and does not help the reader understand the code.	
	Average (4-5)	The documentation is simply comments embedded in the code with some simple header comments separating routines	
	Good (6-8)	The documentation consists of embedded comment and some simple header documentation that is somewhat useful in understanding the code.	
	Outstanding (9-10)	The documentation is well written and clearly explains what the code is accomplishing and how.	
Timely submission (Computer Program)	Poor (0-3)	The code was more than 2 weeks overdue.	
	Average (4-5)	The code was within 2 weeks of the due date.	
	Good (6-8)	The program was delivered within a week of the due date.	
	Outstanding (9-10)	The program was delivered on time	
Efficiency (Computer Program)	Poor (0-3)	The code is huge and appears to be patched together.	
	Average (4-5)	The code is brute force and unnecessarily long	
	Good (6-8)	The code is fairly efficient without sacrificing readability and understanding.	
	Outstanding (9-10)	The code is extremely efficient without sacrificing readability and understanding	

Experiment No. 01

Title: Implement Conflation algorithm to generate document representative of a text file.

Aim:-

To study and implement Conflation Algorithm using File Handling in C/C++.

Objective:-

Find internal Representation for documents in a collection. To study how Conflation algorithm is useful in IR system.

Theory:

Before a computerized information retrieval system can actually operate to retrieve some information, that information must have already been stored inside the computer. Originally it will usually have been in the form of documents. The computer, however, is not likely to have stored the complete text of each document in the natural language in which it was written. It will have, instead, a document representative which may have been produced from the documents either manually or automatically.

The starting point of the text analysis process may be the complete document text, an abstract, the title only, or perhaps a list of words only. From it the process must produce a document representative in a form which the computer can handle.

Luhn's ideas

In one of Luhn's early papers he states: 'It is here proposed that the frequency of word occurrence in an article furnishes a useful measurement of word significance. It is further proposed that the relative position within a sentence of words having given values of significance furnish a useful measurement for determining the significance of sentences. The significance factor of a sentence will therefore be based on a combination of these two measurements.' This quote fairly summarizes Luhn's contribution to automatic text analysis. His assumption is that frequency data can be used to extract words and sentences to represent a document.

Generating document representatives - conflation

Ultimately one would like to develop a text processing system which by means of computable methods with the minimum of human intervention will generate from the input text (full text, abstract, or title) a document representative adequate for use in an automatic retrieval system.

A document will be indexed by a name if one of its *significant* words occurs as a member of that class.

Such a system will usually consist of three parts:

- (1) removal of high frequency words,
- (2) suffix stripping,
- (3) detecting equivalent stems.

The removal of high frequency words, 'stop' words or 'fluff' words is one way of implementing Luhn's upper cut-off. This is normally done by comparing the input text with a 'stop list' of words which are to be removed. Table 2.1 gives a portion of such a list, and demonstrates the kind of words that are involved. The advantages of the process are not only that non-significant words are removed and will therefore not interfere during retrieval, but also that the size of the total document file can be reduced by between 30 and 50 per cent. The second stage, suffix stripping, is more complicated. A standard approach is to have a complete list of suffixes and to remove the longest possible one.

Unfortunately, context free removal leads to a significant error rate. For example, we may well want UAL removed from FACTUAL but not from EQUAL. To avoid erroneously removing suffixes, context rules are devised so that a suffix will be removed only if the context is right. 'Right' may mean a number of things:

- (1) the length of remaining stem exceeds a given number; the default is usually 2;
- (2) the stem-ending satisfies a certain condition, e.g. does not end with Q.

TABLE 1.1 : List of 'Stop' words

A	CANNOT	INTO	OUR	THUS
ABOUT	CO	IS	OURS	TO
ABOVE TOGETHER	COULD	IT	OURSELVES	
ACROSS	DOWN	ITS	OUT	TOO
AFTER	DURING	ITSELF	OVER	TOWARD
AFTERWARDS	EACH	LAST	OWN	TOWARDS
AGAIN	EG	LATTER	PER	UNDER
AGAINST	EITHER	LATTERLY	PERHAPS	UNTIL
ALL	ELSE	LEAST	RATHER	UP
ALMOST	ELSEWHERE	LESS	SAME	UPON
ALONE	ENOUGH	LTD	SEEM	US
ALONG	ETC	MANY	SEEMED	VERY
ALREADY	EVEN	MAY	SEEMING	VIA
ALSO	EVER	ME	SEEMS	WAS
ALTHOUGH	EVERY	MEANWHILE	SEVERAL	WE
ALWAYS	EVERYONE	MIGHT	SHE	WELL
AMONG	EVERYTHING	MORE	SHOULD	WERE
AMONGST	EVERYWHERE	MOREOVER	SINCE	WHAT
AN WHATEVER	EXCEPT	MOST	SO	
AND	FEW	MOSTLY	SOME	WHEN
ANOTHER	FIRST	MUCH	SOMEHOW	WHENCE
ANY WHENEVER	FOR	MUST	SOMEONE	

ANYHOW	FORMER	MY	SOMETHING	WHERE
ANYONE WHEREAFTER	FORMERLY	MYSELF	SOMETIME	
ANYTHING	FROM	NAMELY	SOMETIMES	WHEREAS
ANYWHERE	FURTHER	NEITHER	SOMEWHERE	WHEREBY
ARE	HAD	NEVER	STILL	WHEREIN
AROUND WHEREUPON	HAS	NEVERTHELESS	SUCH	
AS WHEREVER	HAVE	NEXT	THAN	
AT	HE	NO	THAT	WHETHER
BE	HENCE	NOBODY	THE	WHITHER
BECAME	HER	NONE	THEIR	WHICH
BECAUSE	HERE	NOONE	THEM	WHILE
BECOME	HEREAFTER	NOR	THEMSELVES	WHO
BECOMES	HEREBY	NOT	THEN	WHOEVER
BECOMING	HEREIN	NOTHING	THENCE	WHOLE
BEEN	HEREUPON	NOW	THERE	WHOM
BEFORE	HERS	NOWHERE	THEREAFTER	WHOSE
BEFOREHAND	HERSELF	OF	THEREBY	WHY
BEHIND	HIM	OFF	THEREFORE	WILL
BEING	HIMSELF	OFTEN	THEREIN	WITH
BELOW	HIS	ON	THEREUPON	WITHIN
BESIDE	HOW	ONCE	THESE	WITHOUT
BESIDES	HOWEVER	ONE	THEY	WOULD
BETWEEN	I	ONLY	THIS	YET
BEYOND	IE	ONTO	THOSE	YOU

BOTH	IF	OR	THOUGH	YOUR
BUT	IN	OTHER	THROUGH	YOURS
BY YOURSELF	INC	OTHERS	THROUGHOUT	
CAN YOURSELVES	INDEED	OTHERWISE	THRU	

TABLE 2.2 :List of Suffix

ABILITIES	ALISES	ANCIAL	ARISABILITY	
ASISINGFUL				
ABILITY	ALISING	ANCIALS	ARISABLE	
ASISINGLY				
ABLE	ALISINGFUL	ANCIES	ARISATION	
ASISINGS				
ABLED	ALISINGLY	ANCING	ARISATIONS	
ASIZABLE				
ABLEDLY	ALISINGS	ANCINGFUL	ARISE	ASIZE
ABLENESS	ALISM	ANCINGLY	ARISED	ASIZED
ABLENESSES	ALISMS	ANCINGS	ARISEDLY	
ASIZEDLY				
ABLER	ALIST	ANCY	ARISER	ASIZER
ABLES	ALISTIC	ANEOUS	ARISES	ASIZES
ABLING	ALISTICALLY	ANEOUSLY	ARISING	ASIZING
ABLINGFUL	ALISTICISM	ANEOUSNESS	ARISINGFUL	
ASIZINGFUL				
ABLINGLY	ALISTICISMS	ANT	ARISINGLY	
ASIZINGLY				
ABLY	ALISTICS	ANTANEOUS	ARISINGS	
ASIZINGS				
ACEOUS	ALISTS	ANTANEOUSLY	ARISM	ASM
ACEOUSLY	ALITIES	ANTED	ARISMS	ASMS
ACEOUSNESS	ALITY	ANTEDLY	ARIST	AST
ACEOUSNESSES	ALIZATION	ANTIALITIES	ARISTIC	ASTIC
ACIES	ALIZATIONAL	ANTIALITY	ARISTICISM	ASTICAL
ACIDOUS	ALIZATIONALLY	ANTIALNESS	ARISTICISMS	
ASTICALLY				
ACIDOUSLY	ALIZATIONS	ANTIALNESSES	ARISTICS	
ASTICISM				
ACIOUSNESS	ALIZE	ANTIC	ARISTS	
ASTICISMS				
ACIOUSNESSES	ALIZED	ANTICISM	ARITIES	ASTICS
ACITIES	ALIZEDLY	ANTICISMS	ARITY	ASTMENT
ACITY	ALIZER	ANTICS	ARIZABILITIES	
ASTMENTS				
ACY	ALIZES	ANTING	ARIZABILITY	ASTRIES

AE	ALIZING	ANTINGFUL	ARIZABLE	ASTRY
AGE	ALIZINGFUL	ANTINGLY	ARIZATION	ASTS
AGED	ALIZINGLY	ANTINGS	ARIZATIONS	ASY
AGEDLY	ALIZINGS	ANTLY	ARIZE	ATA
AGER	ALLED	ANTMENT	ARISED	
ATABILITIES				
AGES	ALLEDLY	ANTMENTS	ARISEDLY	
ATABILITY				
AGING	ALLIC	ANTRESS	ARIZER	ATABLE
AGINGFUL	ALLICALLY	ANTRESSES	ARIZES	ATABLES
AGINGLY	ALLICISM	ANTRY	ARIZING	ATABLY
AIC	ALLICISMS	ANTS	ARIZINGFUL	ATAL
AICAL	ALLICS	AR	ARIZINGLY	ATE
AICALLY	ALLING	ARIAL	ARIZINGS	ATED
AICALS	ALLINGFUL	ARIALS	ARLY	ATEDLY
AICISM	ALLINGLY	ARIAN	AROID	ATELY
AICISMS	ALLMENT	ARIANS	AROIDS	ATENESS
AICS	ALLY	ARIC	ARS	
ATENESSES				
AL	ALMENT	ARICISM	ARY	ATER
ALISATION	ALNESS	ARICISMS	ASIS	ATES
ALISATIONAL	ALNESSES	ARICS	ASISE	ATIC
ALISATIONALLY	ALS	ARIES	ASISEABLE	ATICAL
ALISATIONS	ANCE	ARILINESS	ASISED	
ATICALLY				
ALISE	ANCED	ARILY	ASISEDLY	ATICISM
ALISED	ANCEDLY	ARINESS	ASISER	
ATICISMS				
ALISEDLY	ANCER	ARINESSES	ASISES	ATICS
ALISER	ANCES	ARISABILITIES	ASISING	ATING

The final output from a conflation algorithm is a set of classes, one for each stem detected. A class name is assigned to a document if and only if one of its members occurs as a significant Automatic text analysis 13 word in the text of the document. A document representative then becomes a list of class names. These are often referred to as the documents *index terms* or *keywords*.

Input:

File containing input text.

Output: The source file contents after REMOVING STOP WORDS

The file contents after SUFFIX STEMMING

Frequency Count

Conclusion: The document is processed and conflated file i.e. document representative is generated.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	What is mean by Information Retrieval?	CO1	Level 1
Q2.	Draw the diagram of IR?	CO1	Level 2
Q3.	Difference between DR and IR	CO1	Level 3
Q4.	Explain Conflation Algorithm	CO2	Level 2
Q5.	What is the necessity of feedback mechanism in IR?	CO2	Level 1
Q6.	What is document representative?	CO2	Level 1
Q7.	List out some high frequency words and why we called them high frequency words?	CO1	Level 1
Q8.	If input string = "some text is here" then what will be the output of Conflation Algorithm?	CO1	Level 3

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	If your searched is based on word which will occur very rarely in the document then what will happen?	CO1	Level 4

Guidelines for Submission:

- ✚ The journal must contain the following related to every experiment:
- ✚ Theory
- ✚ Diagrams wherever necessary
- ✚ Code
- ✚ Input and Output
- ✚ Conclusion

Assessment Methodology:

- ✚ Assessment of submission by student for every experiment will be a continuous assessment for the TW
- ✚ Assessment will be based on understanding of theory, attentiveness during practical, and understanding.
- ✚ Timely submission of journal.

References:

1. Yates & Neto, "Modern Information Retrieval", Pearson Education.
2. C.J. Rijsbergen, "Information Retrieval", (www.dcs.gla.ac.uk).

Experiment No. 02

Title: Implement Single-pass Algorithm for clustering of files.

Aim:-

Implementation of Single-pass Algorithm for Clustering in c/java language.

Objective:-

- ❖ Find internal Representation for documents in a collection. • To study how clustering is useful in IR system to Search for relevant documents and Various Clustering algorithms.

Theory:

Clustering: Process of grouping set of objects into classes of similar objects.
Cluster : A cluster is an ordered list of objects, which have some common characteristics. Cluster Hypothesis.

- ❖ Concept of 'Threshold' and 'Cluster Representative'.
- ❖ Order of the objects entering the system where they are to be clustered in one pass.
- ❖ Compute the centroid of every object and compare it with Cluster representative.
- ❖ If it falls within the threshold then the object enters the particular cluster else creates its own cluster.
- ❖ Repeat the procedure till all the objects are exhausted.

Clustering Algorithm

Type 1: Hierarchical methods , Non-hierarchical methods

Type 2:

- The methods producing exclusive clusters.
- The methods producing overlapping clusters

Clustering in Information Retrieval:

1. Find internal Representation for documents in a collection.
2. Define Clusters with Cluster representatives.
3. Accept user query. Find internal representation of query.
4. Search for relevant documents
 - i. Level 1 : Comparing query and cluster representative.
 - ii. Level 2 : Comparing query and actual document.
5. Display resultant list of documents to user.

Basic Definitions

•Cluster :

A cluster is an ordered list of objects, which have some common characteristics.

•Distance between two clusters

The distance between two clusters involves some or all elements of the two clusters. The clustering method determines how the distance should be computed

•Similarity

A similarity measure $SIMILAR(d_i, d_j)$ can be used to represent the similarity between the documents.

Threshold

The lowest possible input value of similarity required to join two objects in one cluster.

•Similarity matrix

Similarity between objects calculated by the function $SIMILAR(d_i, d_j)$ represented in the form of a matrix is called a similarity matrix.

•Cluster representative (seed)

The representative of the cluster. Every incoming object's similarity is compared with cluster representative.

Similarity between documents

•Similarity: Similarity value indicates how much two documents or objects are near to each other.

•Example:

–X : The list of index terms related to a document e.g. Document 1

–Y : The list of index terms related to a document e.g. Document 2

–| X | : The number of index terms present in the Document 1.

–| Y | : The number of index terms present in the Document 2

–: Intersection of two sets.

Properties of Clustering Algorithm for Information Retrieval

•**Theoretical soundness :**

The clustering method should satisfy some constraints like :

–The method produces a clustering which is unlikely to be altered drastically when further objects are incorporated i.e. it is stable under growth.

–The method is stable in the sense that small errors in the description of the objects lead to small changes in clustering.

–The method is independent of the initial ordering of the objects.

•**Efficiency :**

–The method should be efficient in terms of speed requirement and storage requirement

Single Pass Algorithm : Steps

1.The object descriptions are processed serially

2.The first object becomes the cluster representative of the first cluster.

3.Each subsequent object is matched against all cluster representatives existing at its processing time.

4.A given object is assigned to one cluster (or more if overlap is allowed) according to some condition on the matching function.

5.when an object is assigned to a cluster the representative for that cluster is recomputed

6.If an object fails a certain test it becomes the cluster representative of a new cluster

Implementation of Single Pass Algorithm

Data Structures:

1.float sM[MAX][MAX]={0}; //Similarity Matrix

2.float Treshold; //Threshold

3.int Cluster[MAX][MAX]={0}; //Clusters

1.Cluster[i][0]= number of entries in cluster

2.Cluster[i][1] = Cluster representative

3.From next entry members of the cluster

4.Int count; //No. of valid clusters

Functions:

1.void process_non();

2.void overlap();

3.void calcr(int);

4.Void display();

Analysis of Algorithm

•Time Complexity:

–Worst Case: $\Omega(n^2)$

–Best Case : $\Omega(n)$

–Average Case : $\Omega(n^2)$

Analysis of Algorithm

•Advantage :

–Simple for implementation.

–Less time requirement for searching

•**Disadvantage :**

–The output depends on the sequence in which objects are handled

Input:

Enter the no of Tokens

Enter the no of Documents

Enter the Threshold

Enter the document token matrix

Output:

Number of clusters with their cluster representatives.

Example:

Suppose that we have the following set of documents and terms, and that we are interested in clustering the terms using the single pass method (note that the same method can be used to cluster the documents, but in that case, we would be using the document vectors (rows) rather than the term vector (columns)).

	T1	T2	T3	T4	T5
Doc1	1	2	0	0	1
Doc2	3	1	2	3	0
Doc3	3	0	0	0	1
Doc4	2	1	0	3	0
Doc5	2	2	1	5	1

Start with T1 in a cluster by itself, say C1. At this point, C1 contains only one item, T1, so the centroid of C1 is simply the vector for T1:

$C1 = \langle 1, 3, 3, 2, 2 \rangle$.

Now compare (i.e., measure similarities) of the next item (T2) to centroids of all existing clusters. At this point we have only one cluster, C1 (we will use dot product for simplicity):

$$\text{SIM}(T2, C1) = 1*2 + 1*3 + 0*3 + 1*2 + 2*2 = 11$$

Now we need a pre-specified similarity threshold. Let's say that our threshold is 10. This means that if the similarity of T2 to the cluster centroid is ≥ 10 , then we add T2 to the cluster, otherwise we use T2 to start a new cluster.

In this case. $\text{SIM}(T2, C1) = 11 > 10$. Therefore we add T2 to cluster C1.

We now need to compute the new centroid for C1 (which now contains T1 and T2). The centroid (which is the average vector for T1 and T2 is:

$$C1 = \langle 3/2, 4/2, 3/2, 4/2 \rangle$$

Now, we move to the next item, T3. Again, there is only one cluster, C1, so we only need to compare T3 with C1 centroid. The dot product of T3 and the above centroid is:

$$\text{SIM}(T3, C1) = 0 + 8/2 + 0 + 0 + 4/2 = 6$$

This time, T3 does not pass the threshold test (the similarity is less than 10). Therefore, we use T3 to start a new cluster, C2. Now we have two clusters

$$C1 = \{T1, T2\}$$

$$C2 = \{T3\}$$

We move to the next unclustered item, T4. Since we now have two clusters, we need to compute the MAX similarity of T4 to the 2 cluster centroids (note that the centroid of cluster C2 right now is just the vector for T3):

$$\begin{aligned}\text{SIM}(T4, C1) &= \langle 0, 3, 0, 3, 5 \rangle \cdot \langle 3/2, 4/2, 3/2, 3/2, 4/2 \rangle \\ &= 0 + 12/2 + 0 + 9/2 + 20/2 = 20.5\end{aligned}$$

$$\begin{aligned}\text{SIM}(T4, C2) &= \langle 0, 3, 0, 3, 5 \rangle \cdot \langle 0, 2, 0, 0, 1 \rangle \\ &= 0 + 6 + 0 + 0 + 5 = 11\end{aligned}$$

Note that both similarity scores pass the threshold (10), however, we pick the MAX, and therefore, T4 will be added to cluster C1. Now we have the following:

$$\begin{aligned}C1 &= \{T1, T2, T4\} \\ C2 &= \{T3\}\end{aligned}$$

The centroid for C2 is still just the vector for T3:

$$C2 = \langle 0, 2, 0, 0, 1 \rangle$$

and the new centroid for C1 is now:

$$C1 = \langle 3/3, 7/3, 3/3, 6/3, 9/3 \rangle$$

The only item left unclustered is T5. We compute its similarity to the centroids of existing clusters:

$$\begin{aligned}\text{SIM}(T5, C1) &= \langle 1, 0, 1, 0, 1 \rangle \cdot \langle 3/3, 7/3, 3/3, 6/3, 9/3 \rangle \\ &= 3/3 + 0 + 3/3 + 0 + 9/3 = 5\end{aligned}$$

$$\begin{aligned}\text{SIM}(T5, C2) &= \langle 1, 0, 1, 0, 1 \rangle \cdot \langle 0, 2, 0, 0, 1 \rangle \\ &= 0 + 0 + 0 + 0 + 1 = 1\end{aligned}$$

Neither of these similarity values pass the threshold. Therefore, T5 will

have to go into a new cluster C3. There are no more unclustered items, so we are done (after making a single pass through the items). The final clusters are:

$C1 = \{T1, T2, T4\}$

$C2 = \{T3\}$

$C3 = \{T5\}$

Note: Obviously, the results for this method are highly dependent on the similarity threshold that is used. You should use your judgment in setting this threshold so that you are left with a reasonable number of clusters.

Input: Data about no of documents, no of terms, occurrence of terms in these documents and threshold value in the Matrix form.

Output: Terms are clustered in different clusters as per exclusive and overlapping approach.

Conclusion: Term based Single pass Clustering implemented and understands the clustering with respect to multiple documents.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	What do you meant by cluster? What is cluster Seed?	CO2	Level 1
Q2.	Why we need cluster in IR? What is document based and Term based Clustering?	CO2	Level 2
Q3.	How can we define in relevance between documents?	CO2	Level 3
Q4.	How can we form the cluster of documents?	CO2	Level 2
Q5.	How many methods are there to define cluster?	CO2	Level 1
Q6.	Explain Single pass algorithm.What do you mean bu overlapping and exclusive approach?	CO2	Level 1
Q7.	What is meant by similarity matrix?	CO2	Level 1
Q8.	What is the time complexity and space complexity for Single Pass?	CO2	Level 3
Q9.	Which method is better for cluster definition?	CO2	Level 1

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	Illustrate the situation when clustering will not possible?	CO2	Level 4
Q2.	Create the document based clustering for 5 documents.	CO2	Level 6

Guidelines for Submission:

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- ✚ Diagrams wherever necessary
- ✚ Code
- ✚ Input and Output
- ✚ Conclusion

Assessment Methodology:

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References:

1. Yates & Neto, "Modern Information Retrieval", Pearson Education.
2. C.J. Rijsbergen, "Information Retrieval", (www.dcs.gla.ac.uk).

Experiment No. 03

Title: Implement a program for retrieval of documents using inverted files.

Aim:-

To understand the concept of inverted file and implement a program for Retrieval of documents using inverted files.

Objective:-

To study Indexing , Inverted Files and searching with the help of inverted file in **Java Language**

Theory:

An *inverted file* is a file structure in which every list contains only one record. Remember that a list is defined with respect to a keyword K , so every K -list contains only one record. This implies that the directory will be such that $n_i = h_i$ for all i , that is, the number of records containing K_i will equal the number of K_i -lists. So the directory will have an address for each record containing K_i . For document retrieval this means that given a keyword we can immediately locate the addresses of all the documents containing that keyword. For the previous example let us assume that a non-black entry in the field corresponding to an attribute indicates the presence of a keyword and a black entry its absence. Then the directory will point to the file in the way shown in *Figure 4.3*. The definition of an inverted files does *not* require that the addresses in the directory are in any order. However, to facilitate operations such as conjunction ('and') and disjunction ('or') on any two inverted lists, the addresses are normally kept in record number order. This means that 'and' and 'or' operations can be performed with one pass through both lists. The penalty we pay is of course that the inverted file becomes slower to update.

Indexing

In searching for a basics query is to scan the text sequentially. Sequential or online text searching involves finding the occurrences of a pattern in a text. Online searching is appropriate then the text is small and it is the only choice if the text collection is very volatile or the index space overhead cannot be afforded. A second option is to build data structures over the text to speed up the search. It

is worthwhile building and maintaining an. index when the text collection is large and semi-static. Semi-static collections can be updated at reasonably regular intervals but they are not deemed to support thousands of insertions of single words per second. This is the case for most real text databases not only dictionaries or other slow growing literary works. There are many indexing Techniques.

Three of them are inverted files, suffix arrays and signature files.

Inverted Files:

An inverted file is a word-oriented mechanism for indexing a text collection in order to speed up the matching task. The inverted file structure is composed of two elements: vocabulary and occurrence. The vocabulary is the set of all different words in the text. For each such word a list of all the text portions where the word appears is stored. The set of all those lists is called the occurrences. These positions can refer to words or characters. Word positions simplify phrase and proximity queries, while character positions facilitate direct access to the matching text position.

Searching with the help of inverted file:

The search algorithm on an inverted index has three steps.

1. Vocabulary Search

It includes a description of

Indexing

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Searching with the help of inverted file:

The search algorithm on an inverted index has three steps.

1. Vocabulary Search
2. Retrieval of occurrence
3. Manipulation of occurrences

Single-word queries can be searched using any suitable data structure to speed up the search, such as hashing, tries, or B-trees. The first two give $O(m)$ search cost. However, simply storing the words in lexicographically order is cheaper in space and very competitive in performance. Since the word can be binary searched at $O(\log n)$ cost. Prefix and range queries can also be solved with binary search, tries, or B-trees but not with hashing. If the query is formed by single words then the process ends by delivering the list of occurrences. Context queries are more difficult to solve with inverted indices. Each element must be searched separately and a list generated for each one. Then, the lists of all elements are traversed in synchronization to *find* places where all the words appear in sequence (for a phrase) or appear close enough (for proximity). If one list is much shorter than the others, it may be better to binary search its elements into the longer lists instead of performing a linear merge. If block addressing is used it is necessary to traverse the blocks for these queries, since the position information is needed. It is

then better to intersect the lists to obtain the blocks which contain all the searched words and then sequentially search the context query in those blocks. Some care has to be exercised at block boundaries, since they can split a match.

Example:

Text:

1 6 9 11 17 19

This is a text. A text has many words. Words are made from letters.

Inverted Index:

Vocabulary	Occurrences
Letters	60...
Made	50...
Many	28...
Text	11,19...
Words	33,40....

Example:

D1

. d b c d
1 2 3 4 5

D2

. ; . ; p . d ; d
1 2 3 4 5 6 7 8 9

. and ; signifies stop words which we will not going to consider.

Table 1 for D1

V1/Term	PO1	Freq
d	2,5	2
b	3	1
c	4	1

T

Table 2 for D2

V1/Term	PO2	Freq
p	5	1
d	7,9	2

Table 3:

Term	DOC #
d	1
b	1
c	1
p	2
d	2

Table 4:

Term	DOC #
b	1
c	1
d	1
d	2
p	2

Table 5:

Term	DOC #	Freq
b	1	1
c	1	1
d	1	2
d	2	2
p	2	1

Table 6:Dictionary

Term	N Docs	Total Freq
b	1	1
c	1	1
d	2	4
p	1	1

Table 7:Postings (from table 5)

DOC #	position	Freq
1	3	1
1	4	1
1	2,5	2
2	7,9	2
2	5	1

Steps:

- 1)Input two documents as D1 and D2
- 2)Generate Table1 and table 2 for D1 and D2 respectively.
- 3)Generate table 3as shown above.
- 4)Sort terms alphabetically as shown in table 4
- 5)Merging Multiple entries in single documents (if any) and Merging of single entry in multiplae document (if any) as shown in table 5.
- 6)Genetarete dictionary as shown in table 6
- 7)Generate postings as shown in table 7.
- 8)Input query in triplet format from posting
- 9)Display query result and all the tables.

Above steps can also be summarized as follows

Algorithm
1. Input the conflated file or documents 2. Build the index file for input file 3. Input the query(from table 7 as (1,1,3) .It should give you ‘b’ 4. Print all the tables ,the index file and result of query

Input: Documents containing vocabulary and query in triplet format.

Output: Posting and dictionary and Specific Document as per the query

Conclusion: Concept of inverted file is implemented and understands the storage concept.Also Implemented Algorithm to search the document and evaluate performance is implemented which gives the expected document as per the query.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	What do you mean by inverted file?	CO2	Level 1
Q2.	What is word oriented mechanism?	CO2	Level 1
Q3.	What do you mean by dictionary?	CO2	Level 1
Q4.	What is the role of postings?	CO2	Level 1
Q5.	What are different applications of inverted index?	CO2	Level 1
Q6.	How inverted index is helpful in searching?	CO2	Level 2
Q7.	What do you mean by merging similar terms from different documents.	CO2	Level 1
Q8.	What is the time complexity and space complexity for Single Pass?	CO2	Level 1
Q9.	What are applications of inverted file?	CO2	Level 2

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	Illustrate the situation when inverted file will not be effective as storage option?	CO2	Level 4

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2. C.J. Rijsbergen, "Information Retrieval", (www.dcs.gla.ac.uk).

Experiment No. 04

Title: Implement a program to calculate precision and recall for sample input. (Answer set A, Query q1, Relevant documents to query q1- Rq1)

Aim:-

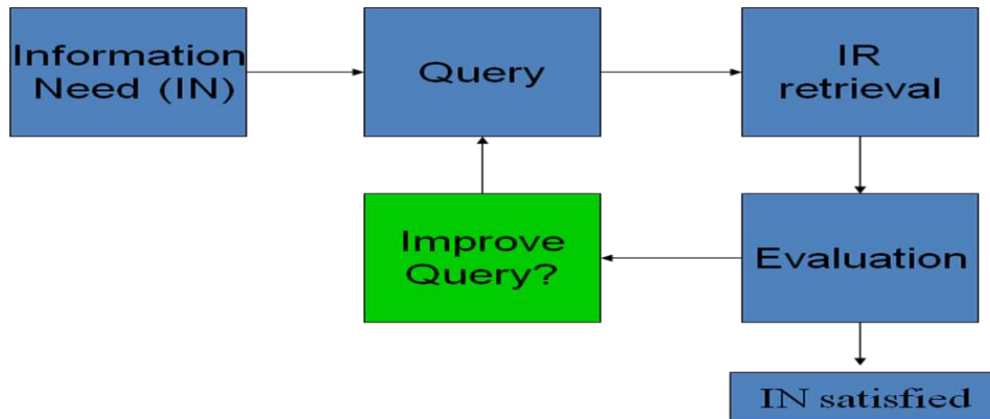
To implement Program to calculate precision and recall for sample input.
(Answer set A, Query q1, Relevant documents to query q1- Rq1)

Objective:-

To study Program to calculate precision and recall for sample input.
(Answer set A, Query q1, Relevant documents to query q1- Rq1)

Theory:

Evaluation Workflow



Relevance as a Measure

- Relevance is everything!
- How relevant is the document
 - for this user for this information need.
- Subjective, but assumes measurable
- Measurable to some extent
 - How often do people agree a document is relevant to a query
- How well does it answer the question?
 - Complete answer? Partial?
 - Background Information?
 - Hints for further exploration?

Relevance

In what ways can a document be relevant to a query?

- **Simple - query word or phrase is in the document.**
 - Problems?
- Answer precise question precisely.
- Partially answer question.
- Suggest a source for more information.
- Give background information.
- Remind the user of other knowledge.
- Others ...

What to Evaluate?

What can be measured that reflects users' ability to use system?

- Coverage of Information
- Form of Presentation
- Effort required/Ease of Use
- Time and Space Efficiency

Effectiveness is through the following:

- Recall
 - proportion of relevant material actually retrieved
- Precision
 - proportion of retrieved material actually relevant

• Recall

$$R = \frac{\text{Number of relevant items retrieved}}{\text{Total number of relevant items in collection}}$$

• Precision

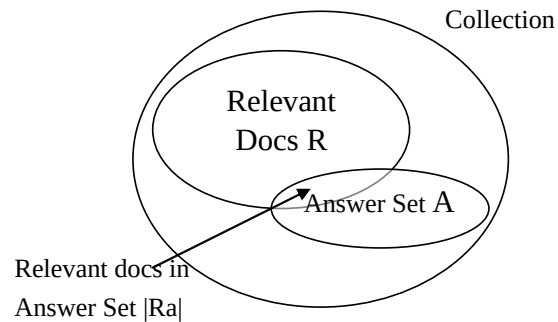
$$P = \frac{\text{Number of relevant items retrieved}}{\text{Total number of items retrieved}}$$

• *Goal*

high recall and high precision

$$\text{Recall} = \frac{|Ra|}{|R|}$$

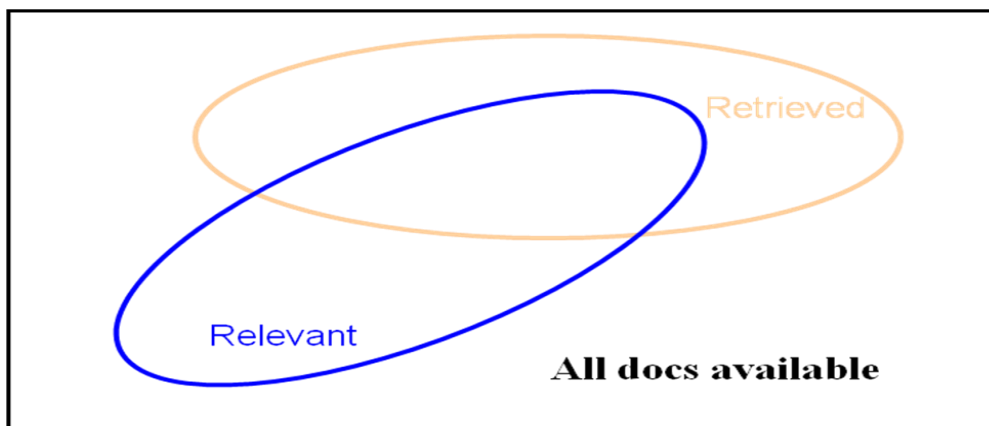
$$\text{Precision} = \frac{|Ra|}{|A|}$$



How do we measure relevance?

- Binary measure
 - 1 relevant
 - 0 not relevant
- N-ary measure
 - 3 very relevant
 - 2 relevant
 - 1 barely relevant
 - 0 not relevant
- N=? consistency vs. expressiveness tradeoff

Relevant vs. Retrieved Documents



Contingency table of relevant and retrieved documents

	Retrieved	Not retrieved	
Relevant	w	x	Relevant = w + x
Not relevant	y	z	Not Relevant = y + z

Retrieved = w + y Not Retrieved = x + z

Total # of documents available N = w + x + y + z

- Precision: $P = w / \text{Retrieved} = w / (w + y)$ $P = [0,1]$
- Recall: $R = w / \text{Relevant} = w / (w + x)$ $R = [0,1]$

Example

- Documents available:
D1,D2,D3,D4,D5,D6,D7,D8,D9,D10
- Relevant to our need:
D1, D4, D5, D8, D10
- Query to search engine retrieves:
D2, D4, D5, D6, D8, D9

	retrieved	not retrieved
relevant	D4,D5,D8	D1,D10

not relevant	D2,D6,D9	D3,D7
--------------	----------	-------

Precision and Recall – Contingency Table

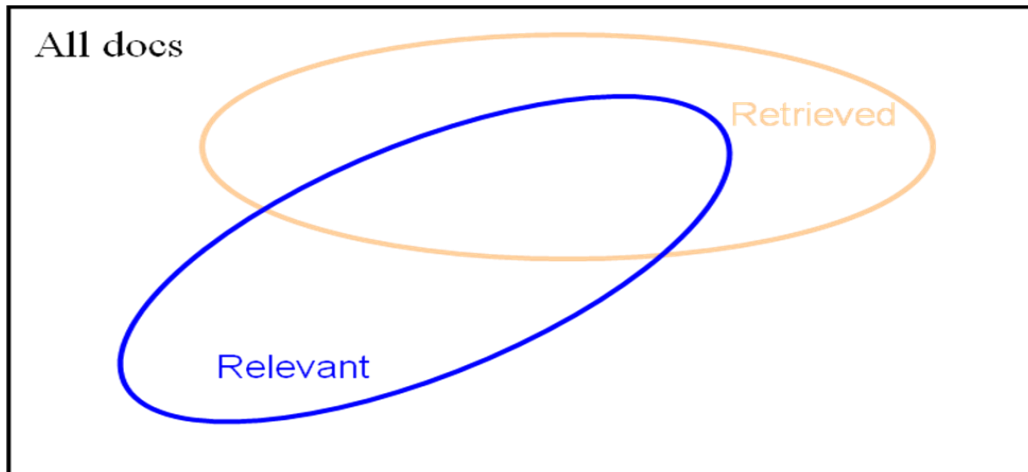
	Retrieved	Not retrieved	
Relevant	w=3	x=2	Relevant = w+x= 5
Not relevant	y=3	z=2	Not Relevant = y+z = 5
	Retrieved = w+y = 6	Not Retrieved = x+z = 4	
	Total documents N = w+x+y+z = 10		

- Precision: $P = w / w+y = 3/6 = .5$
- Recall: $R = w / w+x = 3/5 = .6$

What do we want

- Find everything relevant – high recall
- Only retrieve those – high precision

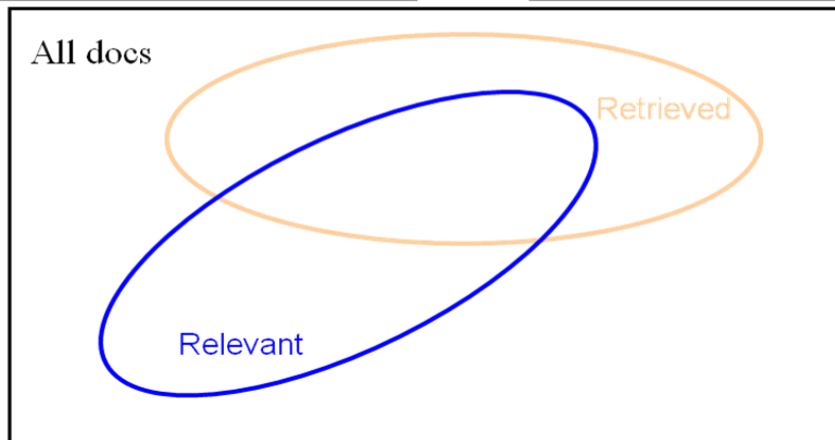
Relevant vs. Retrieved



Precision vs. Recall

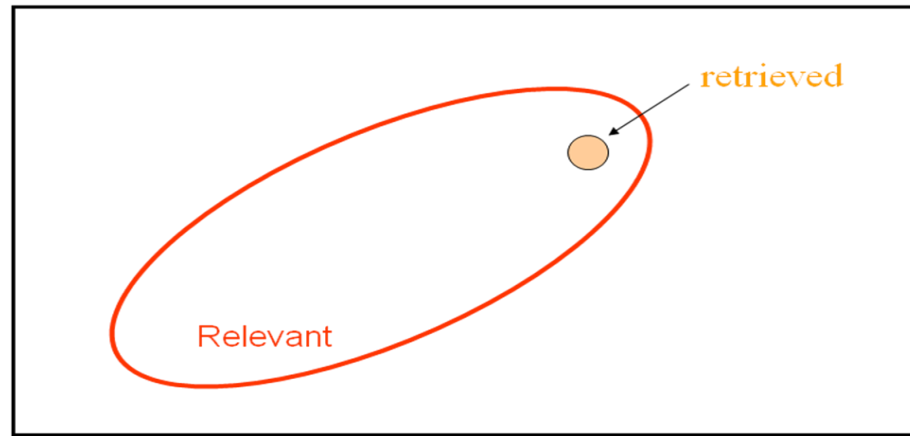
$$\text{Precision} = \frac{|\text{RelRetrieved}|}{|\text{Retrieved}|}$$

$$\text{Recall} = \frac{|\text{RelRetrieved}|}{|\text{Rel in Collection}|}$$

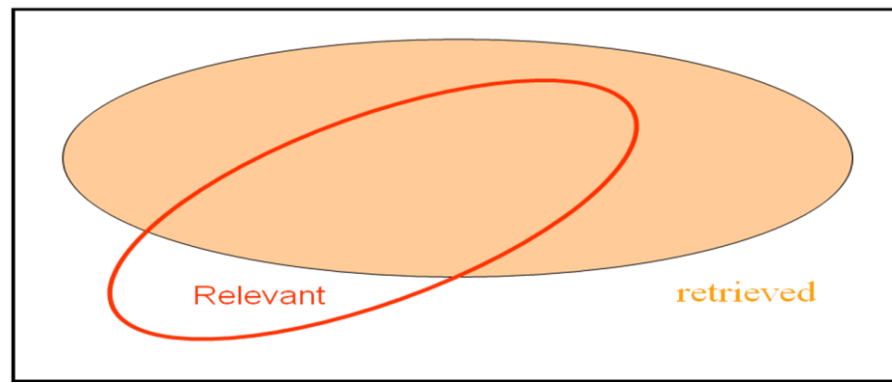


Retrieved vs. Relevant Documents

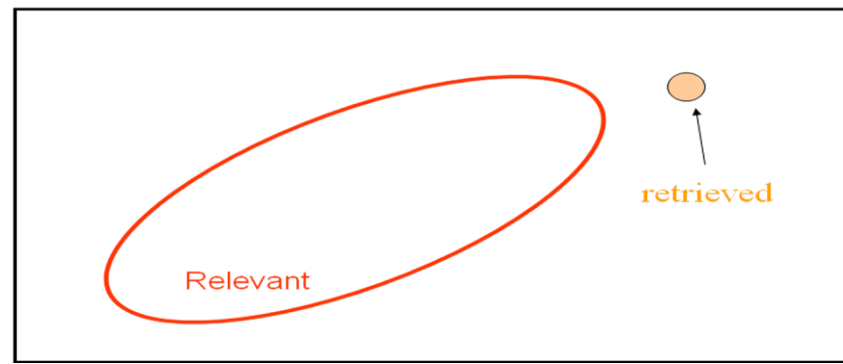
Very high precision, very low recall



High recall, but low precision



Very low precision, very low recall (0 for both)



High precision, high recall (at last!)



Conclusion: Concept of precision and recall implemented to evaluate IR systems.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	What do you mean by Precision?	CO3	Level 1
Q2.	What do you mean by Recall?	CO3	Level 1
Q3.	Explain is the need of performance evaluation of IR system	CO3	Level 2
Q4.	Explain how to evaluate performance of IR system.	CO3	Level 5

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References:

1. C.J. Rijsbergen, "Information Retrieval", (www.dcs.gla.ac.uk).
2. V. S. Subrahmanian, Satish K. Tripathi, "Multimedia information System", Kulwer Academic Publisher.

Experiment No. 05

Title: Write a program to calculate harmonic mean (F-measure) and E-measure for above example

Aim:- To implement to calculate harmonic mean (F-measure) and E-measure for above example

Objective:- To study Program to calculate harmonic mean (F-measure) and E-measure for above example

Theory:

Consider the above example discussed in experiment no 4 and use to calculate harmonic mean (F-measure) and E-measure.

The F-Measure: It is also called as F1 score. The F-measure of the system is defined as **the weighted harmonic mean of its precision and recall**.
Combine Precision and Recall into one number

$$F = \frac{2}{1/R + 1/P} = 2 \frac{RP}{R + P}$$

$P = \text{precision}$

$R = \text{recall}$

$F = [0,1]$

$F = 1$; when all ranked documents are relevant

$F = 0$; no relevant documents have been retrieved

Or

F1-score is the Harmonic mean of the Precision and Recall

It can be calculated as:

$$F_1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

F1-score is a better metric when there are imbalanced classes. It is needed when you want to seek a balance between Precision and Recall.

In most real-life classification problems, imbalanced class distribution exists and thus F1-score is a better metric to evaluate the model under consideration.

The E-Measure: Evaluation measures for an information retrieval system are **used to assess how well the search results satisfied the user's query intent.**

-Combine Precision and Recall into one number

$$E = 1 - \frac{1 + b^2}{\frac{b^2}{R} + \frac{1}{P}}$$

$$E = 1 - \frac{1}{\alpha \left(\frac{1}{P} \right) + (1 - \alpha) \frac{1}{R}}$$
$$\alpha = 1/(\beta^2 + 1)$$

$P = \text{precision}$

$R = \text{recall}$

b = measure of relative importance of P or R

For example,

$b = 0.5$ means user is twice as interested in precision as recall

Conclusion: Concept of F Measure and E measure implemented to evaluate IR systems.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	What do you mean F Measure?	CO3	Level 1
Q2.	What do you mean E Measure?	CO3	Level 1
Q3.	What is the Significance of F and E measure in IR system?	CO3	Level 2

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Experiment No. 06

Title: Implement a program for feature extraction in 2D color images (any features like color, texture etc. and to extract features from input image and plot histogram for the features.

Aim:- To implement Program for Feature Extraction from 2D Colour Images using java

Objective:- To study Program for Feature Extraction from 2D Colour Images for features like Colour and Textures .

Theory:

Color is a widely used important feature for image representation. This is very important as it is invariant with respect to scaling, translation and rotation of an image. Color space, color quantification and similarity measurement are the key components of color feature extraction.

The **RGB color model** is an additive color model in which red, green, and blue light are added together in various ways to reproduce a broad array of colors. The name of the model comes from the initials of the three additive primary colors, red, green, and blue.

The main purpose of the RGB color model is for the sensing, representation, and display of images in electronic systems, such as televisions and computers, though it has also been used in conventional photography. Before the electronic age, the RGB color model already had a solid theory behind it, based in human perception of colors.

RGB is a *device-dependent* color model: different devices detect or reproduce a given RGB value differently, since the color elements (such as phosphors or dyes) and their response to the individual R, G, and B levels vary from manufacturer to manufacturer, or even in the same device over time. Thus an RGB value does not define the same *color* across devices without some kind of color management.

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Color Histogram

A histogram is the distribution of the number of pixels for an image¹. The number of elements in a histogram depends on the number of bits in each pixel of an image.

An **image histogram** is a type of histogram that acts as a graphical representation of the tonal distribution in a digital image.^[1] It plots the number of pixels for each tonal value. By looking at the histogram for a specific image a viewer will be able to judge the entire tonal distribution at a glance.

HISTOGRAM BASICS

A histogram is a graph counting how many pixels are at each level between black and white. Black is on the left. White is on the right. The height of the graph at each point depends on how many pixels are that bright. Lighter images move the graph to the right. Darker ones move it to the left. Easy!

More Pixels



Fewer Pixels



Black - Dark - Medium - Light - White

Histogram

A good image often, but not always, has a histogram spread all over.

Input: Color Image (e.g. .bmp)

Output: color feature extraction and Histogram for it.

Note: Students should also work on another features of color image.

Conclusion: Feature extraction performed and extracted color content. Also studied the Color models.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1	What do you mean by image histogram?	CO4	Level 1
Q2	What are different types of file format for images?	CO4	Level 1
Q3	Explain is the role of color model	CO4	Level 2
Q4.	Compare different color models.	CO4	Level 5
Q5.	Define image histogram?	CO4	Level 1
Q6.	List types of histogram	CO4	Level 2
Q7.	Explain use of image histogram	CO4	Level 1
Q8.	How to extract R ,G and B content from image	CO4	Level 2

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	Consider the image processing domain and explain various methods of extracting intensity of pixel.	CO1	Level 2

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2. V. S. Subrahmanian, Satish K. Tripathi, "Multimedia information System", Kulwer Academic Publisher.

Experiment No. 07

Title: Build the web crawler to pull product information and links from an e-commerce website using Python programming language.

Aim:-

To understand use of Web crawler in searching and Implement Web Crawler in Python

Objective:-

To understand and implement Web crawler which is used by the search engine to index Web pages across the web. To study crawling of the HTML Page to find the keywords on that page for search engine indexing of the pages .

Theory:-

A **Web crawler** is a computer program that browses the World Wide Web in a methodical, automated manner or in an orderly fashion.

A crawler is a program that visits Web sites and reads their pages and other information in order to create entries for a search engine index. The major search engines on the Web all have such a program, which is also known as a "spider" or a "bot." Crawlers are

typically programmed to visit sites that have been submitted by their owners as new or updated. Entire sites or specific pages can be selectively visited and indexed. Crawlers apparently gained the name because they crawl through a site a page at a time, following the links to other pages on the site until all pages have been read.

Other terms for Web crawlers are ants, automatic indexers, bots, Web spiders, Web robots, or especially in the FOAF community - Web scutters.

A web crawler starts with a list of URLs to visit, called the seed. For each URL, the crawler finds links in the HTML, filters those links based on some criteria and adds the new links to a queue. All the HTML or some specific information is extracted to be processed by a different pipeline.

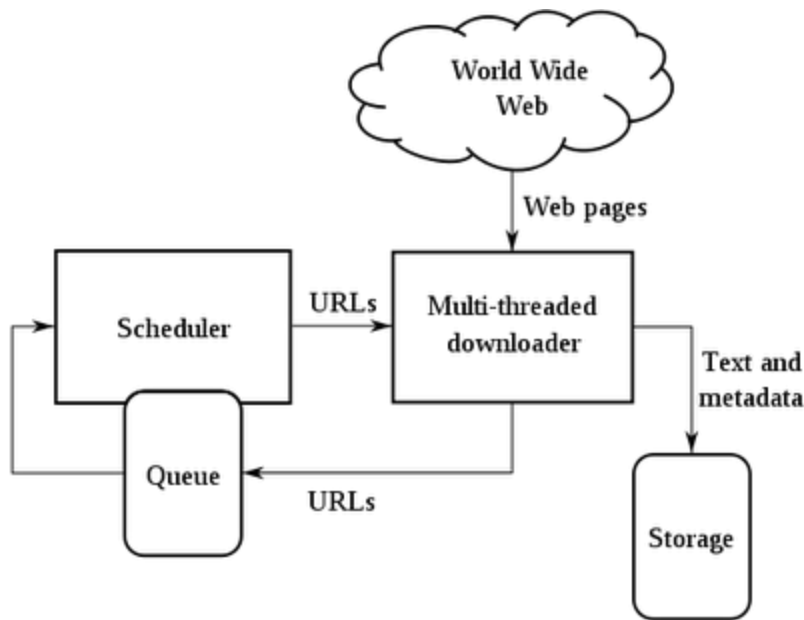
This process is called Web crawling or spidering. Many sites, in particular search engines, use spidering as a means of providing up-to-date data. Web crawlers are mainly used to create a copy of all the visited pages for later processing by a search engine that will index the downloaded pages to provide fast searches. Crawlers can also be used for automating maintenance tasks on a Web site, such as checking links or validating HTML code. Also, crawlers can be used to gather specific types of information from Web pages, such as harvesting e-mail addresses (usually for sending spam).

A Web crawler is one type of bot, or software agent. In general, it starts with a list of URLs to visit, called the seeds. As the crawler visits these URLs, it identifies all the hyperlinks in the page and adds them to the list of URLs to visit, called the crawl frontier. URLs from the frontier are recursively visited according to a set of policies.

The large volume implies that the crawler can only download a fraction of the Web pages within a given time, so it needs to prioritize its downloads. The high rate of change implies that the pages might have already been updated or even deleted.

The number of possible crawlable URLs being generated by server-side software has also made it difficult for web crawlers to avoid retrieving duplicate content. Endless combinations of HTTP GET(URL-based) parameters exist, of which only a small selection will actually return unique content. For example, a simple online photo gallery may offer three options to users, as specified through HTTP GET parameters in the URL. If there exist four ways to sort images, three choices of thumbnail size, two file formats, and an option to disable user-provided content, then the same set of content can be accessed with 48 different URLs, all of which may be linked on the site.

This mathematical combination creates a problem for crawlers, as they must sort through endless combinations of relatively minor scripted changes in order to retrieve unique content.



Web crawler is the program of piece of code that **search engine** uses to index **Web pages across the web**. It crawls the HTML Page to find the **keywords** on that page for search engine indexing of the pages .

Algorithm

1. Make User Interface
2. Input the URL of any website
3. Establish HTTP connection
4. Read HTML page source code
5. Extract Hyperlinks of HTML page
6. Display the list of hyperlinks on the same page

Output

List of Hyperlinks

Pseudo Code for Web Crawler :

Get the user's input: the starting URL and the desired file type. Add the URL to the currently empty list of URLs to search. While the list of URLs to search is not empty,


```

{
  Get the first URL in the list.
  Move the URL to the list of URLs already searched.
  Check the URL to make sure its protocol is HTTP
    (if not, break out of the loop, back to "While").
  See whether there's a robots.txt file at this site
    that includes a "Disallow" statement.
    (If so, break out of the loop, back to "While".)
  Try to "open" the URL (that is, retrieve
    that document From the Web).
  If it's not an HTML file, break out of the loop,
    back to "While."
  Step through the HTML file. While the HTML text
    contains another link,
  {
    Validate the link's URL and make sure robots are
    allowed (just as in the outer loop).
    If it's an HTML file,
      If the URL isn't present in either the to-search
      list or the already-searched list, add it to
      the to-search list.
      Else if it's the type of the file the user
      requested,
        Add it to the list of files found.
    }
  }
}

```

Input: Website name or link

Output: List of URL for that link.

Conclusion: Implemented Web Crawler in python and understand the use of crawling in searching

Frequently Asked Questions:

Que.	Question	CO	BT
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Q1.	What is meant by crawler	C05	Level 1
Q2.	Explain working of web crawler	C05	Level 2
Q3.	Give examples of Web Crawlers.	C05	Level 1
Q4.	Specify role of crawler in searching	C05	Level 3
Q5.	Explain Meta Crawler	C05	Level 2

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	Modify developed WebCrawler considering other parameters or keywords	C05	Level 3

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- ✚ Input and Output
- ✚ Conclusion

Assessment Methodology:

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- ✚ Timely submission of journal.

References:

1. Yates & Neto, "Modern Information Retrieval", Pearson Education.

Experiment No. 08

Title:- Write a program to find the live weather report (temperature, wind speed, description, and weather) of a given city using Python.

Aim:- To implement a program in Python to find the live weather report (temperature, wind speed, description, and weather) of a given city.

Objective:- To Study about the live weather report generation (temperature, wind speed, description, and weather) of a given city in Python

Theory:

Weather Report: a systematic statement of the existing and usually the predicted meteorological conditions over a particular area.

Weather is the combination of four factors—**temperature, wind, precipitation, and sunlight and clouds**—that occur at a given place and time.

Weather warnings are important forecasts because **they are used to protect life and property**. Forecasts based on temperature and precipitation are important to agriculture, and therefore to traders within commodity markets. Temperature forecasts are used by utility companies to estimate demand over coming days

How do I get weather report in Python?

Get Weather Information using Python

1. 1 What is OpenWeatherMap.

2. 2 Prerequisites & Configuration. 2.1 Get an API Key. 2.2 Install Requests.
3. 3 Creating a Simple Weather Showing Python App.
4. 4 Creating a Simple Flask Page for Showing Weather.

Find current weather of any city using OpenWeatherMap API in Python

we are going to get the weather of a city using **OpenWeatherMap** API. To use the OpenWeatherMap API, we have to get the API key. We will get it by creating an account on their website.

Create an account and get your API Key. It's free until 60 calls per minute. You have to pay if you want more than that. For this tutorial, the free version is enough. We need **requests** module for the HTTP requests and **JSON** module to work with the response. Follow the below steps to the weather of any city.

- Import the requests and JSON modules.
- Initialize the base URL of the weather API <https://api.openweathermap.org/data/2.5/weather?>.
- Initialize the city and API key.
- Update the base URL with the API key and city name.
- Send a get request using the requests.get() method.
- And extract the weather info using the **JSON** module from the response.
- # importing requests and json
- import requests, json
- # base URL
- BASE_URL = "https://api.openweathermap.org/data/2.5/weather?"
- # City Name CITY = "Hyderabad"
- # API key API_KEY = "Your API Key"
- # upadting the URL
- URL = BASE_URL + "q=" + CITY + "&appid=" + API_KEY
- # HTTP request
- response = requests.get(URL)
- # checking the status code of the request
- if response.status_code == 200:
- # getting data in the json format
- data = response.json()
- # getting the main dict block
- main = data['main']
- # getting temperature
- temperature = main['temp']
- # getting the humidity
- humidity = main['humidity']
- # getting the pressure
- pressure = main['pressure']
- # weather report
- report = data['weather']

- `print(f'{CITY:-^30}')`
- `print(f'Temperature: {temperature}')`
- `print(f'Humidity: {humidity}')`
- `print(f'Pressure: {pressure}')`
- `print(f'Weather Report: {report[0]['description']}')`
- `else:`
- `# showing the error message`
- `print("Error in the HTTP request")`
- **Output**
- If you run the above program, you will get the following results.
- -----Hyderabad-----
- Temperature: 295.39
- Humidity: 83
- Pressure: 1019
- Weather Report: mist

Another Python Code:

```
import requests
from pprint import pprint
def weather_data(query):
    res=requests.get('http://api.openweathermap.org/data/2.5/weather?'+query+'&AP
PID=*****8&units=metric');
    return res.json();
def print_weather(result,city):
    print("{}'s temperature: {}°C ".format(city,result['main']['temp']))
    print("Wind speed: {} m/s".format(result['wind']['speed']))
    print("Description: {}".format(result['weather'][0]['description']))
    print("Weather: {}".format(result['weather'][0]['main']))
def main():
    city=input('Enter the city:')
    print()
    try:
        query='q='+city;
        w_data=weather_data(query);
        print_weather(w_data, city)
        print()
    except:
        print('City name not found...')
if __name__=='__main__':
    main()
```

Output:

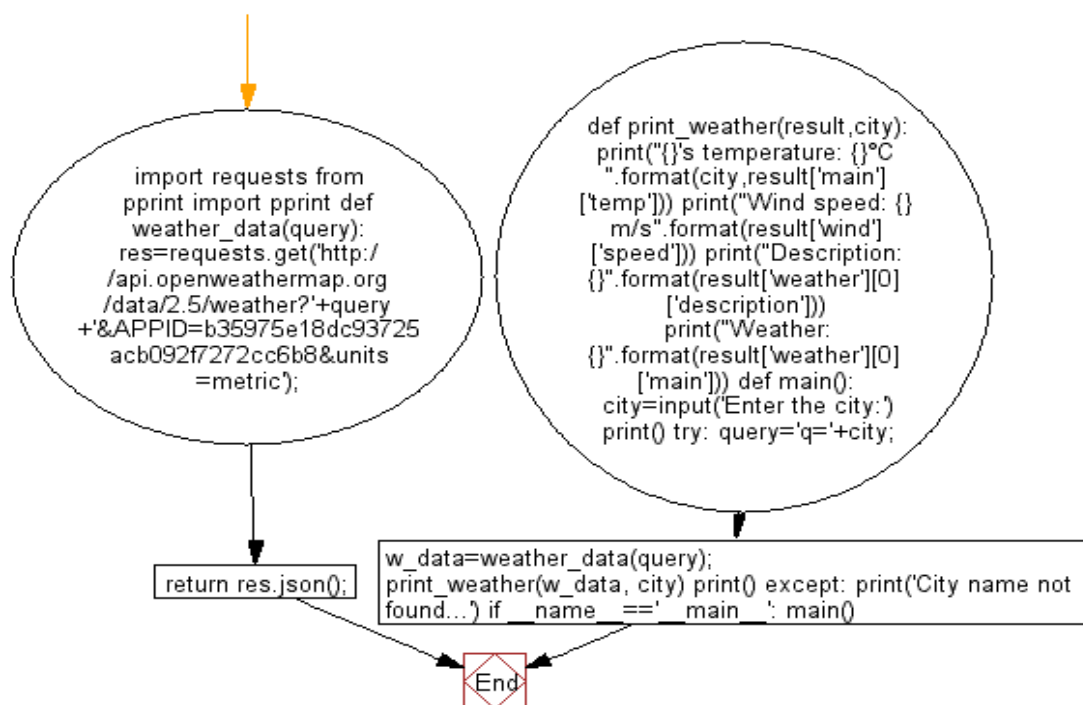
Enter the city: Pune

Pune's temperature: 26.45°C

Wind speed: 2.1 m/s

Description: clear sky

Weather: Clear

Flowchart:

Input: city name

Output: weather information.

Conclusion: Live weather information obtained through Python.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	Explain the steps to implement this assignment?	CO5	Level 1
Q2.	What is OpenWeatherMap service?	CO5	Level 1
Q3.	What is the use of JSON in this assignment?	CO5	Level 2

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q2.	How to use this python code to predict the weather for next few days?	CO5	Level 2

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- ✚ Timely submission of journal.

References:

1. Ryan Mitchell, Web Scraping with Python, O'reilly

Experiment No. 09

Title: Case study on recommender system for a **product** / Doctor / Product price / Music.

Aim:-
Study the need and various types of recommender systems such as Collaborative or content based recommendation system.

Objective:-
To study features and working of collaborative filtering recommender system

Theory:-

Recommendation systems (RS) help to match users with items

- Ease information overload
- Sales assistance (guidance, advisory, persuasion,...)

RS are software agents that elicit the interests and preferences of individual consumers and make recommendations accordingly.

They have the potential to support and improve the quality of the decisions consumers make while searching for and selecting products online.

Different system designs / paradigms

- Based on availability of exploitable data
- Implicit and explicit user feedback
- Domain characteristics

Purpose and success criteria :

- **Different perspectives/aspects**
 - Depends on domain and purpose
 - No holistic evaluation scenario exists
- **Retrieval perspective**
 - Reduce search costs
 - Provide "correct" proposals
 - Users know in advance what they want
- **Recommendation perspective**
 - Serendipity – identify items from the Long Tail
 - Users did not know about existence

Recommender systems

- **RS seen as a function**
- **Given:**
 - User model (e.g. ratings, preferences, demographics, situational context)
 - Items (with or without description of item characteristics)
- **Find:**
 - Relevance score. Used for ranking.
- **Relation to Information Retrieval:**
 - IR is finding material [...] of an unstructured nature that satisfies an information need from within large collections.

Paradigms of recommender systems:

Recommender systems reduce information overload by estimating relevance



Types of Recommender Systems:

1. **Collaborative Filtering (CF)** – match ‘like-minded’ people

- E.g. if two people have similar ‘taste’ they can recommend items to each other



2. **Content-Based (CB)** – use personal preferences to match and filter items

- E.g. what sort of books do I like?

3. **Social Software** – the recommendation process is supported but not automated

- E.g. Weblogs provide a medium for recommendation

4. **Social Data Mining** – Mine log data of social activity to learn group preferences

- E.g. web usage mining

Collaborative Filtering (CF)

- Collaborative-based filtering systems can produce recommendations by computing the similarity between a user’s preferences and the preferences of other people
- Such systems do not attempt to analyse or understand the content of the items being recommended
- They are able to suggest new items to user who have similar preferences with others
- The most prominent approach to generate recommendations
 - used by large, commercial e-commerce sites
 - well-understood, various algorithms and variations exist
 - applicable in many domains (book, movies, DVDs, ..)

- **Approach**
 - use the "wisdom of the crowd" to recommend items
- **Basic assumption and idea**
 - Users give ratings to catalog items (implicitly or explicitly)
 - Customers who had similar tastes in the past, will have similar tastes in the future

Pure CF Approaches

- **Input**
 - Only a matrix of given user–item ratings
- **Output types**
 - A (numerical) prediction indicating to what degree the current user will like or dislike a certain item
 - A top-N list of recommended items

Basic mechanism

- A large group of people's preferences are registered
- A subgroup of people is located whose preferences are similar of the user who seeks the recommendation
- An average of the preferences for that group is calculated
- The resulting preference function is used to recommend options to the user who seeks the recommendation
- The concept of similarity needs to be defined in some way

Example user-item matrix

	<i>Item1</i>	<i>Item2</i>	<i>Item3</i>	<i>Item4</i>	<i>Item5</i>
User <i>A</i>	4	4	1	4	3
User <i>B</i>	2	1	4	2	5
User <i>C</i>	3	1	3	2	1
User <i>D</i>	5	4	2		3

What would be the recommendation for user *D*?

Neighbourhood-based algorithms

Three steps

(i) The degree of similarity of the active user and the others in the database is calculated (positive or negative)

(ii) A set of users is chosen as the basis for making the prediction. This is determined based on the degree of similarity and differs from system to system

(iii) The set of users chosen in the previous step is used to make the recommendation. A user with high degree of similarity may be assigned higher weight

Pearson's correlation coefficients

It reflects the degree of linear relationship between two variables and ranges -1 to $+1$

The degree of correlation between an active user *a* and another user *u* is:

$$w_{a,u} = \frac{\sum_{i=1}^n (r_{a,i} - \bar{r}_a)(r_{u,i} - \bar{r}_u)}{\sqrt{\sum_{i=1}^n (r_{a,i} - \bar{r}_a)^2} \sqrt{\sum_{i=1}^n (r_{u,i} - \bar{r}_u)^2}}$$

$\bar{r}_a$: is the average ratings of the active user

$\bar{r}_u$ is the average of another user *u*'s ratings

$w_{a,u}$ the degree of correlation between user *a* and user *u*

$p_{a,i}$ represents the prediction for user *a* on item *i*

Next, the neighbourhood of users based on which the recommendation will be provided is selected

The weighted average of the ratings of the neighbourhood of users for the item of interest is then calculated as follows:

$$p_{a,i} = \bar{r}_a + \frac{\sum_{u=1}^m (r_{u,i} - \bar{r}_u) w_{a,u}}{\sum_{u=1}^m |w_{a,u}|}$$

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Example

	<i>Item1</i>	<i>Item2</i>	<i>Item3</i>	<i>Item4</i>	<i>Item5</i>
User <i>A</i>	4	4	1	4	3
User <i>B</i>	2	1	4	2	5
User <i>C</i>	3	1	3	2	1
User <i>D</i>	5	4	2		3

Using Pearson's correlation coefficients:

$$w_{D,A} = 0.9 \quad w_{D,B} = -0.7 \quad w_{D,C} = 0$$

$$p_{D,Item4} = 4.5$$

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Comparisons of different Recommender Systems:

	Pros	Cons
Collaborative	No knowledge-engineering effort, serendipity of results, learns market segments	Requires some form of rating feedback, cold start for new users and new items
Content-based	No community required, comparison between items possible	Content descriptions necessary, cold start for new users, no surprises
Knowledge-based	Deterministic recommendations, assured quality, no cold-start, can resemble sales dialogue	Knowledge engineering effort to bootstrap, basically static, does not react to short-term trends

A product recommendation system in e-commerce:

A product recommendation is basically a filtering system that seeks to predict and show the items that a user would like to purchase. It may not be entirely accurate, but if it shows you what you like then it is doing its job right.

Recommendation system for e-commerce. In e-commerce recommendation systems, the main objective is the suggestion of products for consumers in calls smart shop windows or recommendation windows. They can appear on the website's home page, on product pages and in the shopping cart.

Recommender systems are used by E-commerce sites to suggest products to their customers. The products can be recommended based on the top overall sellers on a site, based on the demographics of the customer, or based on an analysis of the past buying behavior of the customer as a prediction for future buying behavior.

Product recommendations can be as simple as presenting each new user with a list of best selling items or as complex as using an algorithm that shows each visitor a dynamically updated set of products. "Personalized product recommendation engines represent the future."

Recommender systems are changing from novelties used by a few E-commerce sites, to serious business tools that are re-shaping the world of E-commerce. Many of the largest commerce Web sites are already using recommender systems to help their customers find products to purchase. A recommender system learns from a customer and recommends products that she will find most valuable from among the available products. In this paper we present an explanation of how recommender systems help E-commerce sites increase sales, and analyze six sites that use recommender systems including several sites that use more than one recommender system. Based on the examples, we create a taxonomy of recommender systems, including the interfaces they present to customers, the technologies used to create the recommendations, and the inputs they need from customers. We conclude with ideas for new applications of recommender systems to E-commerce.

Amazon.com

We focus on recommender systems in the book section of Amazon.com. **Customers who Bought:** Like many E-commerce sites, Amazon.com™ (www.amazon.com) is structured with an information page for each book, giving details of the text and purchase information. The Customers who Bought feature is found on the information page for each book in their catalog. It is in fact two separate recommendation lists. The first recommends books frequently purchased by customers who purchased the selected book. The second recommends authors whose books are frequently purchased by customers who purchased works by the author of the selected book.

Eyes: The Eyes feature allows customers to be notified via email of new items added to the Amazon.com catalog. Customers enter requests based upon author, title, subject, ISBN, or publication date information. Customers can use both simple and more complex Boolean-based criteria (AND/OR) for notification queries. Requests can also be directly

entered from any search results screen, creating a persistent request based on the search

Amazon.com Delivers: Amazon.com Delivers is a variation on the Eyes feature. Customers select checkboxes to choose from a list of specific categories/genres (Oprah books, biographies, cooking). Periodically the editors at Amazon.com send email announcements to notify subscribers of their latest recommendations in the subscribed categories. Book Matcher: The Book Matcher feature allows customers to give direct feedback about books they have read. Customers rate books they have read on a 5-point scale from “hated it” to “loved it.” After rating a sample of books, customers may request recommendations for books they might like. At that point a half dozen non-rated texts are presented which correlate with the user’s indicated tastes. Feedback to these recommendations is provided by a “rate these books” feature where customers can indicate a rating for one or more of the recommended books.

Customer Comments: The Customer Comments feature allows customers to receive text recommendations based on the opinions of other customers. Located on the information page for each book is a list of 1-5 star ratings and written comments provided by customers who have read the book in question and submitted a review. Customers have the option of incorporating these recommendations into their purchase decision

Conclusion: Various recommender system studied and every recommender system has benefits under specific circumstances. RS for Product also studied.

Frequently Asked Questions:

Que.	Question	CO	BT
Q1.	Define recommender system	CO6	Level 1
Q2.	List types of recommender system.	CO6	Level 1
Q3.	Explain how the Recommender systems reduce information overload by estimating Relevance ?	CO6	Level 2
Q4.	Explain Collaborative filtering	CO6	Level 2
Q5.	Explain Hybrid Recommender system and	CO6	Level 2

Questions related to the applications of the experiment

Que.	Question	CO	BT
Q1.	If you want to buy a product from Amazon.in website. Explain how recommender system is helpful to enhance your buying product experience.	CO6	Level 2

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-

References:

1. Ricci, F, Rokach, L. Shapira, B.Kantor, “Recommender Systems Handbook”.